Yijing Wu

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EDUCATION

Northeastern University | Seattle, WA, U.S. September 2021 – Present
Master of Science in Information Systems; GPA: 3.9/4.0
- Courses: Web Design and User Experience Engineering, Smartphones-Based Web Development

Tongji University | Shanghai, China September 2017 – June 2021
Bachelor of Engineering in Communication Engineering (Electronic Engineering)
- Courses: Algorithms Design and Analysis, Software Technology, Principles of Database Systems, Computer Communication Networks, Brain and Cognitive Computing, Principles of Artificial Intelligence

SKILLS

- **Programming Languages**: JAVA, JavaScript, TypeScript, Python, HTML, CSS, SQL, C/C++, Swift
- **Tools/Frameworks**: React, React Native, Node.js, Django, Express, MySQL, MongoDB, Vue, Docker, AWS, GCP, Git

WORK EXPERIENCE

Upduo | San Francisco, CA, U.S. June 2022 – Present
Software Engineer Intern (**React, React Native, Django, GraphQL, TypeScript, Python, Java, Objective-C**)
- Worked as a full-stack engineer to build and improve Upduo's web/mobile app via **React, React Native** and **Django**
- Integrated **iOS** framework CallKit and **Android** ConnectionService via Callkeep and modified the native code with **Objective-C** and **Java** to realize a video call notification feature
- Performed local camera and microphone selection and check features through **Agora** to ensure **WebRTC** video quality
- Collaborated with product and design team to conduct visual changes and deliver new user interface, such as animation, time windows, data visualization
- Implemented beta testing and unit tests to locate Upduo web/mobile app issues

Huawei Technologies | Shanghai, China July 2020 – September 2020
Algorithm Engineer Intern (**Python, C, OpenCV, Anaconda**)
- Applied calibration and distortion correction algorithm to process video flows of camera sensor for self-driving vehicles
- Achieved reduction of distortion correction errors from 1.395 ± 0.136 pixels to 0.055 ± 0.192 pixels through cylindrical projection and single-point mapping
- Increased processing efficiency of DSP by developing mapping table and video processing algorithms for digital signal processors based on **OpenCV**

PROJECT EXPERIENCE

Rubik's Cube Helper, Northeastern University (React.js, Node.js, REST API, GCP) November 2021 - March 2022
- Delivered a dynamic and cross-browser compatible web application to bring more personalized Rubik's Cube experience
- Implemented responsive design, accessibility development, and multiple interactions met WCAG 2.1 AA standard

Digital Image Lab, Tongji University (Python, Matlab, EEGLab) September 2019 – June 2021
- Constructed classification models of brain during working memory maintenance based on 128-channel EEG dataset for material-specific Sternberg task through feature selection and machine learning
- Extracted features of EEG signals by empirical mode decomposition (EMD) and phase space reconstruction (PSR)
- Executed Support Vector Machine (SVM), K-nearest neighbor (KNN), Random Forest (RF), and nested cross-validation
- Accomplished improvement of classification performance from 71.45% to 91.82% under SVM and published the result in an IEEE paper as the first author

Positioning and Navigation Lab, Tongji University (Python, Unity3D, EasyAR, C#) April 2019 – June 2021
- Built a Barrier-Free Positioning and Navigation System for people with disabilities by performing image segmentation via **PyTorch** for indoor map construction as well as detecting obstacles based on the **YoloV3** algorithm
- Guided a four-person research team and participated in the development of **AR** assisted module for indoor positioning and navigation system in China International Import Expo (CIIE) in 2019 and 2020

Course Selection Management System, Tongji University (Vue.js, Node.js, MySQL, JS, SQL) June 2020 - June 2020
- Developed a full-stack course selection management system with the separation of frontend and backend
- Devised web page based on **Vue.js** and **Element-UI**, and utilized **Node.js** and **MySQL** for backends to maintain data

ACADEMIC PUBLICATIONS

Classification of EEG signals during Working-Memory Maintenance based on Phase Space Reconstruction of Empirical Mode Decomposition, Oct 2020, ISBN:978-1-6654-2299-4